# SMOOTHNESS PULLS BACK ALONG ETALE MORPHISMS: X -> Z SMOOTH AND Y -> Z ETALE IMPLY X -> Y SMOOTH

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 3960-3964 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 3966-3977, the complete author proof (10 body lines).

*Editorial note.* Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The NL criteria for smooth and etale morphisms (lemma-NL-smooth, lemma-NL-etale, same chapter), the six-term exact sequence (lemma-exact-sequence-NL), and finite presentation permanence (Morphisms); each cited in the retained span.

*Source reference.* Original lemma 0G7Z: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3884–3890. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0D0N: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3862–3869. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0E44: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3916–3926. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 02FV: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3977–3977. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $X \to Y \to Z$ be morphisms of schemes. Assume $X \to Z$ smooth and $Y \to Z$ étale. Then $X \to Y$ is smooth.*

*Proof.* The morphism $X \to Y$ is locally of finite presentation by Morphisms, Lemma 02FV. By Lemma 0D0N we have $H^{-1}(NL_{X/Z}) = 0$ and the module $\Omega_{X/Z}$ is finite locally free. By Lemma 0G7Z we have $H^{-1}(NL_{Y/Z}) = H^0(NL_{Y/Z}) = 0$. By Lemma

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0E44 we get $H^{-1}(NL_{X/Y}) = 0$ and $\Omega_{X/Y} \cong \Omega_{X/Z}$ is finite locally free. By Lemma 0D0N the morphism $X \to Y$ is smooth. $\square$